Are India's schoolteachers prepared to teach?

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TEACHERS are the most important school-based resource for improving student learning. But are schoolteachers in India equipped with the skills they need to be effective inside the classroom? Several indicators suggest otherwise. Low student learning levels are perhaps the most salient indicator, but there are other indicators that imply teachers are not adequately prepared to handle the complex task of teaching and classroom management.

First, there is growing evidence that many teachers lack the content knowledge needed to be effective. In a study of Bihar teachers, average subject knowledge of teachers was found to be weak, with the mean score in language at 45% and math at 58%.[1] Even when teachers displayed content knowledge, many could not explain concepts. For example, almost 80% of teachers had the correct answer to a long division problem (3 digit by 1 digit), but among them only 11% could do all the steps right.

Second, studies on teacher time-on-task suggest that Indian teachers spend most of their teaching time doing exactly those tasks associated with poor teaching: reading aloud, writing on the blackboard and barely engaging with their students.[2] Finally, nearly 27% of students avail of private tuitions at the primary level and more at the secondary level.[3] This is true for government school as well as private school students. Why students take private tuitions is a complex issue, driven by multiple factors – but it is unlikely their learning needs are being met in school if they need to go elsewhere to learn.

When teachers come poorly prepared to teach and manage classrooms, it is important to look at the pre-service education landscape that prepares and allows teacher-candidates to potentially seek a career in teaching. A good pre-service education programme is the first step toward equipping teacher-candidates with the content, pedagogical and managerial skills they might need for becoming effective teachers. That is how it is with other professions. For instance, one cannot be an effective doctor without having good pre-service medical education. Ironically, however, the quantitative evidence on the causal impact of pre-service education on a teacher's ability to improve student learning is inconclusive. Most studies find relatively little impact of a teacher's pre-service education on student learning; furthermore, there is limited consensus on which factors in pre-service education affect student learning more.[4]

Uncertainty regarding the effects of teacher pre-service education comes largely from three methodological challenges.[5] First, it is difficult to isolate teacher productivity, especially since a student's own ability, peer influences, and other characteristics of schools also affect measured outcomes. The problem is exacerbated by the biases resulting from the fact that assignment of students and teachers to classrooms is usually not random.

Second, there is an inherent selection problem in evaluating the effects of education and training on teacher productivity. Unobserved characteristics, such as motivation or intelligence of teacher-candidates, may affect the amount and types of education and training they choose to obtain as well as their subsequent performance in the classroom.

Third, it is difficult to obtain data that provide adequate detail about the various types of training teachers receive and even more difficult to link the training of teachers to the achievement of the students they teach.

Nevertheless, if we look at countries where teaching is a sought after career and student learning is high, a few things stand out about their pre-service education programmes.[6] First, entry into pre-service is highly selective. For instance, becoming a primary schoolteacher is highly competitive in Finland. Selection for primary schoolteacher education happens in two steps: first, candidates are selected based on scores in matriculation exam and out-of-school accomplishment records. Second, candidates take a written exam in pedagogy, their social and communication skills are observed in a clinical setting similar to school situations, and top candidates are interviewed and asked to explain their motivation to become teachers.

About one in every 10 applicants in Finland is accepted in teacher education programmes to become a primary schoolteacher. In Singapore, the government recruits the top one-third of high school graduates to enter teacher education programmes and does not require an entrance exam. In Korea, entrants into teacher education programmes are among the top 10% of high school graduates.

Second, once in pre-service education, candidates follow a curriculum that is linked to what happens in schools, and contain an extensive practical teaching component. In Finland, Korea and Shanghai, a practicum is required for primary and secondary levels comprising at least a six-month classroom teaching component. This allows teacher candidates to learn to apply pedagogical skills and gain skills in classroom management. It acquires special significance as teachers increasingly have to teach a very heterogenous group of students who are at different learning levels.

Third, pre-service education programmes in high performing education systems are closely linked with universities. This allows the curriculum to be informed by the latest research in learning and other fields, while also providing pre-service education a status similar to other undergraduate programmes. Finally, pre-service education programmes in all these countries are tightly regulated by the government. At the primary level, they are generally provided by the government as well.

In India, in contrast, the situation is very different, despite the landmark Justice J.S. Verma Commission report and recommendations presented in 2012. The report provides a clear road map for improving the teacher education system in India. But six years later, pre-service education programmes do not appear to be helping teacher-candidates become effective teachers any more than before.

The poor quality of teachers coming out of teacher training institutes is evident in the low pass rates in teacher eligibility tests. Less than 20% of teachers have passed the test in states where tests have been conducted. This suggests that the majority of those who take the teacher eligibility tests do not qualify to be recruited as teachers despite having the required academic degree and professional training in teacher education.[7]

When comparing India's teacher pre-service education landscape to well performing education systems, none of the pre-conditions are in place in India. First, pre-service education programmes are not selective. There are no entrance tests nor systematic processes for entry into such programmes. To put it bluntly, just about anyone can apply to become a teacher. Second, the curriculum of pre-service education programmes is divorced from the goals of school education as well as the realities of classroom practice.

At both the elementary and secondary level, the curriculum is fragmented and outdated, and does not address subject knowledge adequately. New developments in specific subjects are not incorporated. The focus is on general methods of teaching such as lecture, classroom discussion, question and answer, and memorization. Student teachers do not learn pedagogy skills.[8] Practice teaching in classrooms, for instance, lasts no more than five to six weeks and provides only piecemeal experiences of a fully functioning teacher.

A further concern in India, given the low selectivity of pre-service education programmes, is that teacher-candidates come from the same low quality schools where they will be heading to teach. There is little in the design of pre-service programmes to help remediate the academic deficiencies teacher-candidates bring with them, many of whom come from poor school systems, as they train to become teachers.

Assessments in teacher education programmes are of limited value in screening good candidates from weak ones. The current method of evaluating students enrolled in pre-service teacher education in India, for instance, does not sufficiently assess conceptual and pedagogic skills of teaching.[9] In addition, only cognitive skills are assessed. There is no assessment of student teachers' social and emotional and communication skills to engage with children.

Third, pre-service education colleges in India are typically stand alone, not benefitting from the latest research on teaching or links with other departments. At the primary and secondary levels, three types of teacher training institutes exist: government, privately aided and privately unaided. These include: at the primary level, the District Institutes of Education and Training in each state, which are fully financed by the central government, and at the secondary level, the Regional Institutes of Education that are also central government funded.

In rare cases, universities have departments of education. But for the vast majority, teacher education is located outside the realm of higher education, with most institutes located outside of university campuses.[10] This is often also true of those institutes that are affiliated with universities such as colleges of teacher education. The programmes are isolated, lack well defined professional standards, have low visibility, and do not benefit from new knowledge generated in universities.

A fallout of this is that the institutional capacity to prepare teacher educators is insufficient across the country, with too few programmes offering M.Ed. relative to the needs of specific states. The programmes are general and not able to address specific needs to develop subject experts. There is no policy for professional development for teacher educators.[11]

Finally, weak governance and ineffective regulations characterize the pre-service education landscape in India. There has been a significant increase in the number of teacher education institutes in the past decade. By 2011, the number of teacher education institutes increased to 14704 from about 1800 in 2008.[12] More recent official numbers are not publicly available. Most of this expansion has taken place in the private self-financing sector.[13] The tremendous growth in the numbers of teacher education institutes speaks to the unregulated nature of entry of these institutes: there are considerably more teacher-candidates being prepared than jobs available.

A key problem in managing quality is the relative lack of capacity to enforce norms and standards. This has led to wide variation in quality and a clutter of institutions and programmes in teacher education. As an example, to maintain quality, the National Council for Teacher Education (NCTE) is mandated with inspecting a sample of existing recognized teacher education institutes annually. But it is able to visit only a small number of institutes every year; in 2011 it inspected 168 out of 14704 institutes.[14] Members on the inspection panel may not have the expertise to do the task. This has led to poor quality of inspections and inaccurate decisions by NCTE.

As an example, the Government of Haryana wrote to the National Council for Teacher Education (NCTE) in 2007 to refrain from granting permission to new teacher education institutes as there was no need for them; yet, NCTE granted permission to 114 new colleges in 2009-10, 37 in 2010-11, 19 in 2012-13 and 19 in 2014-15.[15]

Furthermore, inspections are announced in advance to the institutes. When there is an inspection, institutes allegedly put up a show of faculty and students. There have been instances of recognition granted by NCTE in the face of false or fabricated documents and incorrect information submitted by these institutes.[16] There are also numerous news reports that degrees in education can be bought.[17]

In addition to capacity constraints, regulations have been difficult to enforce because several privately run teacher training institutes are owned by powerful vested interests. There is growing concern that most of these institutes are diploma mills driven solely to make profits, where qualifications can be bought easily. This makes implementing the recommendations of the Justice J.S. Verma Commision and cleaning up the pre-service education landscape difficult.

NCTE, under its former Chairperson, Santhosh Mathew, had initiated a process to enforce strict quality control in the system, using third-party auditors. But this process hit roadblocks, which is unsurprising given that many of these institutes were being operated for profit by powerful interests versus qualifying a cadre of motivated teacher-applicants.

If India is serious about improving student learning levels, it must ensure teachers are adequately prepared before they enter the classroom. The Verma Commission report already tells us what needs to be done. But much greater political will is needed to implement these recommendations, including addressing the capacity constraints required to implement the recommendations meaningfully.

Footnotes:

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4. Dan Goldhaber, 'Evidence-Based Teacher Preparation: Policy Context and What We Know', *Journal of Teacher Education* (forthcoming).

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6. Tara Beteille, Michelle Riboud, Shin Nomura, Namrata Tognatta and Yashodhan Ghorpade, 2018, op. cit.

7. MHRD, 'Vision of Teacher Education in India: Quality and Regulatory Perspective.' Report of the Justice Verma Commission. GoI, Delhi, 2012. http://mhrd.gov.in/sites/upload_files/mhrd/files/document-reports/JVC%20Vol%201.pdf

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9. MHRD, 2012, ibid.

10. Ibid.

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12. MHRD, 2012, op. cit; UNESCO, ‘Status of Teacher Education in the Asia-Pacific Region.’ Prepared by International Reading Association, 2008 (all of our countries).

13. MHRD, 2012, op. cit.

14. Ibid.

15. Priyanka Pandey, ‘Pre-Service Education in South Asia.’ Background Note. South Asia Companion to World Development Report, 2018.

16. MHRD, 2012, op.cit.

17. Priyanka Pandey, 2018, op. cit.